# Non-Abelian Clifford $Q_8$ Holonomy, Exact Soldered Laplacians, and the 4D Spin(4) Obstruction on the Homochiral Chair44 Substitution Complex

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**Abstract**

We isolate and prove the core discrete-geometric, spectral, and gauge-theoretic results governing the 3D homochiral `Chair44` substitution tiling and its 4D hypercubic analogue $C_4 = [0,2]^4 \setminus (1,2]^4$. First, we certify Kamenetsky's 8 octant orientation frames (`FR30`) in $\mathrm{SO}(3,\mathbb{Z})$ and confirm one-step coarsening closure $|P_0| = 30 \to |P_1| = 44 = |P_2| = |P^*|$ with $|F^*| = 135$ facet contact triples. Second, we resolve the discrete gauge-theory coboundary problem: whereas any connection constructed solely from relative vertex frames $V_{ab} = M_a^\top M_b$ is a trivial 1-coboundary with identity Wilson loops $W_{\mathrm{vertex}}(\mathcal{C}) = I_3$ on all cycles, the panel-normal Clifford connection $C_{ab}^{(f)} = (\boldsymbol{\sigma}\cdot\mathbf{u}_a)(\boldsymbol{\sigma}\cdot\mathbf{u}_b) \in Q_8$ ($\mathbf{u}_a = M_a^\top \mathbf{n}_f$) depends explicitly on the shared 2-face normal $\mathbf{n}_f$. We prove that $C_{ab}^{(f)}$ transforms covariantly under local spinor gauge transformations $G_a \in Q_8 \subset \mathrm{SU}(2)$, has strictly gauge-invariant Wilson loop traces $\mathrm{Tr}(W_{\mathrm{Clifford}}(\mathcal{C}))$, and generates the full non-abelian quaternion group $\mathrm{Hol}(\nabla_{\mathrm{Clifford}}) = Q_8$ across the 41 fundamental cycles of the Level-1 panel multigraph. Third, we prove by exact integer symbolic factorization that the $8\times 8$ Level-1 tile adjacency Laplacian has characteristic polynomial $\det(\lambda I_8 - L_{\mathrm{tile},1}) = \lambda(\lambda-24)(\lambda^2-30\lambda+207)(\lambda^2-21\lambda+99)^2$ and spectrum $\{0, \frac{21\pm 3\sqrt{5}}{2}, 15\pm 3\sqrt{2}, 24\}$. Fourth, we resolve Kamenetsky's Open Problem 9.1 in $d=4$ by constructing the $64\times 6$ integer boundary bivector obstruction matrix $\mathcal{M}_{\mathrm{obs}}$ on $\partial C_4 \cong S^3$ and certifying $\mathrm{rank}_{\mathbb{Q}}(\mathcal{M}_{\mathrm{obs}}) = 6$ ($\det(\mathcal{M}_{\mathrm{obs}}^\top \mathcal{M}_{\mathrm{obs}}) = 21{,}414{,}742 > 0$). All theorems are accompanied by machine-checked certificates in Lean 4 (`certificates/Chair44Certificates.lean`) and exact integer/symbolic SymPy verification (`src/chair44/machine_checked_certificates.py`).

## 1 Certified `FR30` Orientation Frames and Coarsening Closure

Let $C_3 = [0,2]^3 \setminus (1,2]^3$ denote the 3D $L$-chair polycube composed of 7 unit cubes indexed by $U_3 = \{0,1\}^3 \setminus \{(1,1,1)\}$. Under doubled-coordinate inflation, a parent chair pose $(G,\mathbf{T}) \in \mathrm{SO}(3,\mathbb{Z}) \ltimes \mathbb{Z}^3$ subdivides into 8 child chairs: one central child at translation $\mathbf{T}_0 = (1,1,1)^\top$ with orientation $M_0 = \pi_0$, and 7 octant children indexed by $\mathbf{v} \in U_3$ at translations $\mathbf{T}_{\mathbf{v}} = 4\mathbf{v}$ with orientations

$$M_{\mathbf{v}} = D_s(\mathbf{v})\,\pi_{\mathbf{v}}, \qquad D_s(\mathbf{v}) = \mathrm{diag}(1-2v_1, 1-2v_2, 1-2v_3), \tag{1}$$

where $\pi_{\mathbf{v}} \in S_3$ is a coordinate permutation matrix chosen so that $\det(M_{\mathbf{v}}) = \det(D_s(\mathbf{v}))\det(\pi_{\mathbf{v}}) = +1$.

**Proposition 1.1** (Machine-Checked `FR30` Certification and Pose Closure)**.** *Under Kamenetsky's `FR30` permutation assignment* $\pi_0 = (0,1,2)$, $\pi_{(0,0,0)} = (0,1,2)$, $\pi_{(0,0,1)} = (0,2,1)$, $\pi_{(0,1,0)} = (2,1,0)$, $\pi_{(0,1,1)} = (1,2,0)$, $\pi_{(1,0,0)} = (1,0,2)$, $\pi_{(1,0,1)} = (0,1,2)$, *and* $\pi_{(1,1,0)} = (2,0,1)$, *all 8 Level-1 orientation matrices* $M_0,\ldots,M_7$ *satisfy* $M_a^\top M_a = I_3$ *and* $\det(M_a) = +1$ *over* $\mathbb{Z}$.

*Furthermore, the face-adjacent relative pose language* $(M_u^\top M_v, M_u^\top(\mathbf{T}_v - \mathbf{T}_u))$ *closes in a single coarsening step:*

$$|P_0| = 30 \longrightarrow |P_1| = 44 \longrightarrow |P_2| = 44 = |P^*|, \qquad |F^*| = 135. \tag{2}$$

## 2 Resolution of the Gauge-Theory Coboundary Trap via Face-Normal Clifford Transport

A central mathematical question in discrete gauge theory on polyhedral complexes is why connections built purely from vertex orientation frames have trivial holonomy, and how face-normal soldering escapes this obstruction.

**Theorem 2.1** (Vertex Coboundary Trap vs. Gauge-Invariant Non-Abelian Clifford $Q_8$ Holonomy)**.** *Let* $\mathcal{G}_1 = (V_1, E_{\text{panel}})$ *be the Level-1* `Chair44` *contact multigraph on* $|V_1| = 8$ *child chairs and* $|E_{\text{panel}}| = 48$ *shared unit-square boundary panels.*

1. ***Vertex Coboundary Triviality:*** *For any edge* $e = (a, b)$*, the relative vertex frame connection* $V_{ab} = M_a^\top M_b \in \mathrm{SO}(3, \mathbb{Z})$ *is an exact discrete 1-coboundary. On every closed cycle* $\mathcal{C} = (v_0, v_1, \ldots, v_m = v_0)$*, the path-ordered Wilson loop telescopes identically to the identity:*
$$W_{\text{vertex}}(\mathcal{C}) = M_{v_0}^\top M_{v_1} M_{v_1}^\top M_{v_2} \cdots M_{v_{m-1}}^\top M_{v_0} = I_3. \tag{3}$$

2. ***Face-Normal Clifford*** $Q_8$ ***Connection:*** *Let* $\mathbf{n}_f \in \{\pm\mathbf{e}_1, \pm\mathbf{e}_2, \pm\mathbf{e}_3\}$ *be the ambient unit normal of shared panel* $f$ *directed from tile* $a$ *to tile* $b$*, and let* $\mathbf{u}_a^{(f)} = M_a^\top \mathbf{n}_f$ *and* $\mathbf{u}_b^{(f)} = M_b^\top \mathbf{n}_f$ *be its pullbacks into the local frames of* $a$ *and* $b$*. Define the Clifford transport operator in* $\mathrm{Cl}_{3,0}(\mathbb{R}) \cong M_2(\mathbb{C})$*:*
$$C_{ab}^{(f)} = (\boldsymbol{\sigma}\cdot\mathbf{u}_a^{(f)})(\boldsymbol{\sigma}\cdot\mathbf{u}_b^{(f)}) = (\mathbf{u}_a^{(f)}\cdot\mathbf{u}_b^{(f)})I_2 + i(\mathbf{u}_a^{(f)}\times\mathbf{u}_b^{(f)})\cdot\boldsymbol{\sigma} \in Q_8 = \{\pm I_2, \pm i\sigma_1, \pm i\sigma_2, \pm i\sigma_3\}. \tag{4}$$
*Because* $\mathbf{u}_a^{(f)}$ *depends on the edge-specific panel normal* $\mathbf{n}_f$*,* $C_{ab}^{(f)}$ *does not factor as* $g_a^{-1} g_b$*.*

3. ***Gauge Covariance and Trace Invariance:*** *Under arbitrary local spinor gauge transformations* $\psi_a \mapsto G_a \psi_a$ *with* $G_a \in Q_8$ *(or* $U_a \in \mathrm{SU}(2)$*) and* $C_{ab}^{(f)} \mapsto G_a C_{ab}^{(f)} G_b^{-1}$*, the Wilson loop around any closed cycle* $\mathcal{C}$ *rooted at* $v_0$ *transforms covariantly as* $W_{\text{Clifford}}(\mathcal{C}) \mapsto G_{v_0} W_{\text{Clifford}}(\mathcal{C}) G_{v_0}^{-1}$*, leaving the Wilson loop trace* $\mathrm{Tr}(W_{\text{Clifford}}(\mathcal{C}))$ *strictly gauge-invariant. Moreover, under any global ambient rotation* $R \in \mathrm{SO}(3, \mathbb{Z})$ *acting via* $M_a \mapsto R M_a$ *and* $\mathbf{n}_f \mapsto R\mathbf{n}_f$*, the local vectors* $\mathbf{u}_a^{(f)} = (RM_a)^\top (R\mathbf{n}_f) = M_a^\top \mathbf{n}_f$ *are strictly invariant.*

4. ***Surjectivity onto*** $Q_8$***:*** *Across the* $b_1(\mathcal{G}_1) = 48 - 8 + 1 = 41$ *fundamental cycles of the Level-1 panel multigraph, the holonomy distribution realizes all 8 elements of* $Q_8$*:*
$$\textit{Counts}(Q_8) = \{+1 : 17,\ -1 : 1,\ +i : 6,\ -i : 5,\ +j : 1,\ -j : 5,\ +k : 1,\ -k : 5\}, \tag{5}$$
*with* $24/41$ *(*$58.54\%$*) non-trivial cycles and* $23/41$ *(*$56.10\%$*) non-abelian* $\mathrm{Tr} = 0$ *cycles. On the coarse 8-vertex tile graph (9 fundamental 3-cycles),* $5/9$ *cycles are non-abelian (*$\mathrm{Tr} = 0$*). Thus* $\mathrm{Hol}(\nabla_{\text{Clifford}}) = Q_8$*.*

## 3 Exact Algebraic Spectrum of the Level-1 Adjacency Laplacian

**Theorem 3.1** (Level-1 Tile Adjacency Laplacian Spectrum)**.** *Let $L_{\text{tile},1} = D_{\text{tile},1} - A_{\text{tile},1} \in \mathbb{Z}^{8\times 8}$ be the weighted panel-contact Laplacian of the 8 Level-1 child chairs:*

$$L_{\text{tile},1} = \begin{pmatrix} 21 & -3 & -3 & -3 & -3 & -3 & -3 & -3 \\ -3 & 12 & -3 & -3 & 0 & -3 & 0 & 0 \\ -3 & -3 & 12 & 0 & -3 & 0 & -3 & 0 \\ -3 & -3 & 0 & 12 & -3 & 0 & 0 & -3 \\ -3 & 0 & -3 & -3 & 9 & 0 & 0 & 0 \\ -3 & -3 & 0 & 0 & 0 & 12 & -3 & -3 \\ -3 & 0 & -3 & 0 & 0 & -3 & 9 & 0 \\ -3 & 0 & 0 & -3 & 0 & -3 & 0 & 9 \end{pmatrix}, \qquad \operatorname{Tr}(L_{\text{tile},1}) = 96. \tag{6}$$

*Its characteristic polynomial over $\mathbb{Z}[\lambda]$ factors exactly as:*

$$\det(\lambda I_8 - L_{\text{tile},1}) = \lambda(\lambda - 24)(\lambda^2 - 30\lambda + 207)(\lambda^2 - 21\lambda + 99)^2, \tag{7}$$

*yielding the exact algebraic spectrum (with multiplicities in parentheses):*

$$\operatorname{Spec}(L_{\text{tile},1}) = \left\{ 0^{(1)},\ \left(\frac{21 - 3\sqrt{5}}{2}\right)^{(2)},\ (15 - 3\sqrt{2})^{(1)},\ \left(\frac{21 + 3\sqrt{5}}{2}\right)^{(2)},\ (15 + 3\sqrt{2})^{(1)},\ 24^{(1)} \right\}. \tag{8}$$

**Theorem 3.2** (Exact Schur-Complement Coarsening Decoupling and Soldered Bundle Mass Gap)**.** *Partition the 8 Level-1 child chairs into the central interior chair $I = \{0\}$ ($L_{II} = 21$) and the 7 outer octant chairs $B = \{1, \dots, 7\}$.*

1. ***Exact Radial Mode Decoupling via Schur Complement:*** *Because the central chair shares exactly $w_{0b} = 3$ unit-square panels with every outer chair $b \in B$, the coupling vector is $L_{BI} = -3\mathbf{1}_7$, and the Schur-complement coarsened operator on $B$ is:*

$$L_{\text{eff}} = L_{BB} - L_{BI} L_{II}^{-1} L_{IB} = L_{BB} - \frac{3}{7}\mathbf{1}_7\mathbf{1}_7^\top \in \mathbb{Q}^{7\times 7}, \qquad \operatorname{Tr}(L_{\text{eff}}) = 72. \tag{9}$$

   *Because $-\frac{3}{7}\mathbf{1}_7\mathbf{1}_7^\top$ is a rank-1 projector along $\mathbf{1}_7$ that vanishes on $\mathbf{1}_7^\perp$, Gaussian elimination of the central chair preserves the acoustic zero mode $L_{\text{eff}}\mathbf{1}_7 = \mathbf{0}_7$ and eliminates the radial breathing mode $\lambda = 24$ without shifting any of the remaining 6 algebraic eigenvalues:*

$$\det(\lambda I_7 - L_{\text{eff}}) = \lambda(\lambda^2 - 30\lambda + 207)(\lambda^2 - 21\lambda + 99)^2 = \frac{\det(\lambda I_8 - L_{\text{tile},1})}{\lambda - 24}. \tag{10}$$

2. ***Full-Rank*** $24 \times 24$ ***Soldered Bundle Laplacian:*** *Let $\Delta_{\text{solder},1} \in \mathbb{Z}^{24\times 24}$ be the vector bundle Laplacian with block entries $(\Delta_{\text{solder},1})_{aa} = \sum_{b\neq a} w_{ab}(I_3 + \mathbf{t}_{ab}\mathbf{t}_{ab}^\top)$ and $(\Delta_{\text{solder},1})_{ab} = -w_{ab}M_a^\top M_b$. Then $\Delta_{\text{solder},1}$ is symmetric with exact trace $\operatorname{Tr}(\Delta_{\text{solder},1}) = 384$, and exact rational Gaussian elimination certifies full rank:*

$$\operatorname{rank}_{\mathbb{Q}}(\Delta_{\text{solder},1}) = 24, \qquad \dim\ker(\Delta_{\text{solder},1}) = 0, \qquad \lambda_{\min}(\Delta_{\text{solder},1}) = 2.6687265 > 0, \tag{11}$$

   *proving that non-trivial rotational soldering $M_a^\top M_b \neq I_3$ and translational soldering $\mathbf{t}_{ab}\mathbf{t}_{ab}^\top$ lift all 3 translational zero modes.*

# 4 The 4D Tesseract Chair $C_4$ and Spin(4) Coarsening Obstruction

In $d = 4$, the tesseract chair $C_4 = [0, 2]^4 \backslash (1, 2]^4$ consists of $f_4 = 15$ unit hypercubes $U_4 = \{0, 1\}^4 \backslash \{(1, 1, 1, 1)\}$ with $f_0 = 80$, $f_1 = 212$, $f_2 = 210$, $f_3 = 92$, and $\chi(C_4) = 80 - 212 + 210 - 92 + 15 = 1$ ($\mathbf{b}(C_4) = (1, 0, 0, 0, 0)$). Its closed boundary 3-complex $\partial C_4$ has $f_0^\partial = 80$, $f_1^\partial = 208$, $f_2^\partial = 192$, $f_3^\partial = 64$, $\chi(\partial C_4) = 0$, and Betti vector $\mathbf{b}(\partial C_4) = (1, 0, 0, 1) \cong S^3$.

**Theorem 4.1** (Full-Rank 4D Spin(4) Boundary Obstruction; Resolution of Open Problem 9.1)**.** *Under the chiral Lie algebra splitting* $\mathfrak{spin}(4) \cong \mathfrak{su}(2)_L \oplus \mathfrak{su}(2)_R$ *spanned by self-dual* ($J_i^+$) *and anti-self-dual* ($J_i^-$) *generators* ($[J_a^+, J_b^-] = 0$, $\sum_i (J_i^\pm)^2 = -3I_4$)*, the 8 odd-weight vertices* $\mathbf{v} \in U_4$ ($\sum_\mu v_\mu \equiv 1 \pmod 2$) *satisfy* $\det(D_s(\mathbf{v})) = -1$ *and require an odd permutation* $\pi_\mathbf{v} \in S_4 \setminus A_4$ *to maintain* $\det(M_\mathbf{v}) = +1$*. Every odd permutation in* $S_4$ *couples non-trivially to both* $\mathfrak{su}(2)_L$ *and* $\mathfrak{su}(2)_R$*. Across the 64 boundary 3-cubes of* $\partial C_4$*, the* $64 \times 6$ *integer bivector incidence matrix* $\mathcal{M}_{\text{obs}} \in \mathbb{Z}^{64\times 6}$ *has exact rational rank and Gram determinant:*

$$\operatorname{rank}_\mathbb{Q}(\mathcal{M}_{\text{obs}}) = 6, \qquad \dim\ker(\mathcal{M}_{\text{obs}}) = 0, \qquad \det(\mathcal{M}_{\text{obs}}^\top \mathcal{M}_{\text{obs}}) = 21{,}414{,}742 > 0, \tag{12}$$

*where the exact* $6 \times 6$ *integer Gram matrix is:*

$$\mathcal{M}_{\text{obs}}^\top \mathcal{M}_{\text{obs}} = \begin{pmatrix} 17 & -5 & 0 & 1 & -1 & -3 \\ -5 & 19 & 0 & -1 & -2 & -1 \\ 0 & 0 & 21 & 0 & -2 & -1 \\ 1 & -1 & 0 & 18 & 0 & 0 \\ -1 & -2 & -2 & 0 & 14 & 1 \\ -3 & -1 & -1 & 0 & 1 & 15 \end{pmatrix}. \tag{13}$$

*Consequently, no pure-rotational* $\mathrm{SO}(4, \mathbb{Z})$ *frame assignment on* $C_4$ *can close under doubled-coordinate inflation without parity reflections.*

## Data and Code Availability

All exact integer matrices, SymPy verification scripts, and Lean 4 proof certificates are publicly available in the repository at `certificates/Chair44Certificates.lean`, `src/chair44/machine_checked_ce` and `data/open_audit_package/chair44_fr30_and_laplacian_matrices.json`.